28. THE RENAL SYSTEM : EMBRYONIC KINETICS IN DETAIL

DURATION : 06:06

STUDY SHEET

COURSE SUMMARY

This video deeply explores the development of the renal system, focusing on the embryonic sequence from the pronephros to the metanephros. Starting from day 28, it addresses the key stages of glomerulus formation and the preparation of pro-urine, while highlighting the significant impact of the liver on renal development. In parallel, it discusses the interactions between the Wolffian and Müllerian ducts and the importance of the mesodermal layer in renal constitution. This cutting-edge approach aims to enrich the theoretical and practical understanding of embryonic development, offering students and therapists valuable tools to integrate this knowledge into their osteopathic practice.

THE RENAL SYSTEM: EMBRYONIC KINETICS IN DETAIL

THE PRONEPHROS: FIRST RENAL OUTLINE

The **rapid growth** of the liver leads to the compression and disappearance of the **pronephros**. Despite its regression, the pronephros gives rise to the **mesonephric duct**, a small, primordial collecting duct.

Around **day 28**, this duct forms a primary pronephros, characterised by a small **glomerulus**, the first image of a functional kidney. This is the time of **neural tube closure**, the integration of the **external coelom** into the internal coelom, and the delimitation of the embryo.

PREPARATION OF PRO-URINE

At this stage, the embryo begins to prepare the first **pro-urine**, which will contribute to the increase in **amniotic fluid**.

Initially an exudate from amniotic cells, the amniotic fluid then receives contributions from the **lungs**. This link strengthens around **days 35-40**. It is truly around **day 45**, with the complete closure of the anterior white line, that a first pro-urine appears. An **autocrine exchange** then establishes with the embryonic system, where the embryo is "constellated" by what it will eject and drink.

Day 28 thus marks a first functional connection with the appearance of several **glomeruli** at the level of the mesonephros.

THE MESONEPHROS AND HEPATIC INFLUENCE

The embryo continues to grow, the liver develops more and more, leading to the progressive regression of the **mesonephros**. The upper part of the mesonephric structures is compressed and disappears under the effect of this hepatic growth.

This highlights the importance of the liver in renal development, even for definitive function. This hepatic compression is necessary to subsequently induce the **metanephros**, the definitive renal structure.

The metanephros is already present in the structure of the **intermediate mesoderm**, but not yet induced.

THE WOLFFIAN DUCT AND THE MÜLLERIAN DUCT

The duct called **"Wolffian duct"**, **"mesonephric duct"** or **"ductus mesonephricus"** are synonyms.

In the urogenital system, two major ducts are essential: the Wolffian duct and the Müllerian duct. These two ducts will integrate to form the urogenital system.

The Wolffian duct or ductus mesonephricus induces the metanephros, which corresponds to the terminal part of the intermediate mesoderm, the latter being the entire posterior mesoderm of the embryo.

POSTERIOR DEVELOPMENT AND SWELLING

This posterior mesoderm, corresponding to the **posterior peritoneal cavity**, is the site of a swelling movement. Behind the aorta, a bulge occurs.

This is where another structure appears: the **adrenal glands** (or medullary adrenal glands initially). A small duct is induced here, the **bud** or the **ureteric duct**, which will induce the **metanephric blastema**.

THE SEQUENCE OF RENAL DEVELOPMENT

The embryo possesses an intermediate mesoderm containing all the **mesodermal** potential for the kidney. The growth and delimitation of the embryo organise a flexion movement.

1. **Pronephros**: The posterior part of this mesoderm, by compression of its anterior part, forms a tube, a mesodermal concentration. This process, associated with the approximation of the aortas, induces a rotational movement. From the cervical level, flexion induces a pronephros that segments into small pouches, leaving a small descending duct called the mesonephric duct.

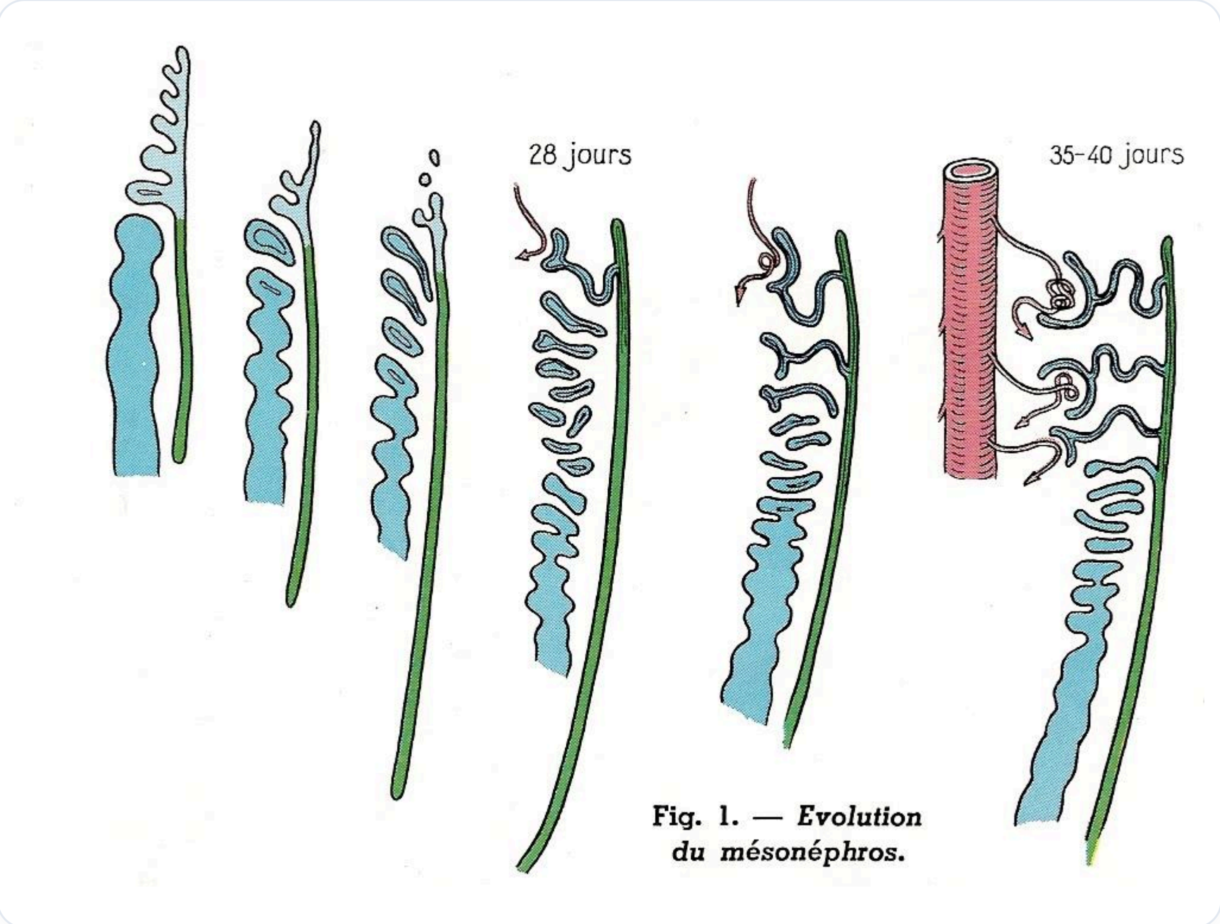

Fig. 1. — *Evolution du mésonéphros.*

FIG. — Evolution of the mesonephros

2. **Mesonephros**: The pronephros disappears very quickly, giving way to a second embryonic renal structure: the mesonephros. A connection is established with the mesonephric duct, and the mesonephros segments into small vesicles to form a first small pronephros. The latter comes into contact with the vascular system around day 28 and begins to function around day 45.

3. **Metanephros**: Then, the growth of the liver and the lengthening of the embryo lead to the disappearance of the mesonephros. Only the collecting duct remains. From this duct, a small opening forms, creating a **ureteric bud** by a suction movement. This bud induces the terminal part of the intermediate mesoderm into a **meso-metanephric blastema**, which will give rise to the definitive kidney.